

Faithfulness Under Paraphrase: Auditing the Consistency of AI Explanations in Text Classifiers

ABSTRACT

Post-hoc feature attribution tools like SHAP and Integrated Gradients are popular methods for opening the black box of text classifiers. Ideally, one would expect that post-hoc tools would be able to identify the reasoning that leads to the prediction and not get distracted by the way the sentence is formulated. However, what about explanations if we rephrase the sentence without changing its meaning or affecting the model's decision? Surprisingly, although the usage of such explainers is quite common, there have been no empirical studies on whether the explanations provided by them are robust under a paraphrase that does not change the meaning of sentence. This issue has been discussed in some survey papers recently. (<https://arxiv.org/abs/2506.21812>)

In an attempt to close this gap, I propose a metric called Faithfulness-Under-Paraphrasing (FUP). Using DistilBERT pre-trained sentiment classifier on standard benchmarks SST2 and IMDB, I am building a paraphrased dataset in which the predicted class label by the classifier is identical. Then, I will use SHAP Partition explainer and Captum Integrated Gradients methods to build token-level attributions and measure the degree of explanation agreement using top-k token overlap (Jaccard similarity) and Spearman's rank correlation. Furthermore, I will examine the impact of several factors on explanation stability, including the choice of an attribution method, degree of rephrasing (edit distance and BLEU scores) and the correctness of the classification.

I expect that both methods will show some degree of instability, with SHAP and Integrated Gradients being unstable in different ways and under different circumstances. Such conclusions will have practical implications for everyone who uses post-hoc explanations to debug models and build user trust. In order to make the research reproducible, I will provide code, prompts and paraphrase data in a public GitHub repository. This work contributes in the following aspects: (1) an intuitive metric for faithfulness grounded in SHAP consistency, (2) the first systematic audit of explanation stability in response to paraphrasing (3) practical guidelines for post-hoc explainability.

Primary RQ: Would feature attribution explanations for the same text classifier using techniques such as SHAP and Integrated Gradients be consistent even after the input text is paraphrased to mean the same thing and produce the same prediction as before? Or would there be significant shifts in the explanations?

RQ1: How sensitive are the SHAP and Integrated Gradients attributions to changes in the original input that maintains the same meaning? (measured using top-k overlap and Spearman rank correlation)

RQ2 : Are SHAP and Integrated Gradients different in terms of their robustness to paraphrasing? For example is one of the methods more prone to degradation, or differently to the other one?

RQ3: What conditions lead to increased instability? Is it affected by the degree of paraphrasing of input relative to the source (edit distance/BLEU), or are there differences between instances where the model is right or wrong?

Keywords: explainable AI, SHAP, Integrated Gradients, faithfulness, robustness, text classification.